A Canonical Correlation of Self-Reward Culture and Financial Control among University Students

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***Abstract*— The growing popularity of self-reward culture among university students has influenced their consumption behavior and financial decision-making. This study aims to examine the relationship between Self-Reward Culture and Financial Control using Canonical Correlation Analysis (CCA). Data were collected from 105 Bina Nusantara University students through a six-point Likert-scale questionnaire designed to measure three dimensions of Self-Reward Culture and three dimensions of Financial Control. Prior to analysis, validity, reliability, normality, outlier, and multicollinearity tests were conducted. Four multivariate outliers were removed using Mahalanobis Distance, and item parceling was applied to satisfy the assumption of multivariate normality. The results showed a significant relationship between Self-Reward Culture and Financial Control, with the first canonical correlation coefficient of 0.663 ($p < 0.001$). Spending Tendency in Self-Reward and Consumptive Behavior were identified as the dominant dimensions contributing to the relationship. The findings suggest that students who engage more frequently in spending-oriented self-reward activities tend to exhibit stronger consumptive tendencies. Therefore, financial education programs should address not only financial literacy but also psychological motivations underlying spending behavior.**

Keywords—Self Reward Culture, Financial Control, Canonical Correlation Analysis, Consumptive Behavior, University Students

I. Introduction

The rapid growth of digital technology, e-commerce platforms, and social media has significantly influenced consumption behavior among university students. One phenomenon that has become increasingly common is self-reward culture, where individuals purchase products or experiences as a reward for achieving personal goals or overcoming challenges [1], [2].

Among students, self-reward is often perceived as a form of self-appreciation and stress relief. However, frequent self-reward activities may lead to excessive spending, Previous studies have shown that self-reward is positively associated with consumptive behavior and impulsive purchasing decisions among Generation Z [2], [8], [9]. Individuals often justify discretionary spending as a reward for their efforts, potentially reducing their ability to prioritize financial needs over wants [8].

Financial control refers to the ability to regulate spending, monitor expenses, maintain budgets, and make responsible financial decisions. Prior research has found that self-control and responsible financial behavior positively contribute to financial well-being and financial management [4], [5], [10].

Despite extensive research on self-reward and consumptive behavior, limited studies specifically examine the impact of self-reward culture on financial control among university students. Therefore, this study aims to investigate the relationship between self-reward culture and financial control.

II. Related Work

Existing studies primarily focuses on how self-reward behavior shapes shopping habits. Rahman et al. [1] found that a self-reward culture, paired with social media, strongly influences Gen Z's purchasing decisions, while Nugroho [2] noted it often serves as a justification for impulsive spending. Additionally, Putri et al. [4] and Pratama and Fauzi [5] identified self-reward as a factor affecting personal finance, though they mixed it with broader variables like self-control.

Other researchers focus on the bigger picture of financial behavior, security, and well-being. Strömbäck et al. [10] found that financial behavior bridges the gap between self-control and financial security.

Despite these insights, previous studies have mostly treated self-reward as a broad driver for shopping, leaving a clear gap regarding its direct link to financial control among university students. Therefore, this study focuses directly on how self-reward culture affects financial control among university students, aiming to provide a clearer understanding of how this mindset influences a student's ability to budget, regulate spending, as well as maintain financial discipline.

III. Theoritical Framework

A. Multivariate Normality Test

Multivariate normality is a key assumption for multivariate analysis, including Confirmatory Factor Analysis (CFA) and Canonical Correlation Analysis (CCA). A dataset meets this assumption if all variables jointly follow a multivariate normal distribution. This is commonly checked using Mardia's multivariate kurtosis coefficient [13], calculated as:

$$b_{2,p} = \frac{1}{n}\sum_{i=1}^{n}[(x_i - \bar{x})'S^{-1}(x_i - \bar{x})]^2$$

Where n is the number of observations, S is the sample covariance matrix, and x_i is the observation vector. A critical ration (CR) falling between -2.58 and 2.58 indicates multivariate normality as a 5% significance level [14].

B. Multicollinearity Test

Multicollinearity happens when independent variables are highly correlated with each other, which can destabilize parameter estimates. To check for this, the Variance Inflation Factor (VIF) is used [15]:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

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where R_j^2 is the coefficient of determination when predictor j is regressed on all remaining predictors. A VIF value under 10 shows that multicollinearity is not an issue [15].

C. Canonical Correlation Analysis (CCA)

Canonical Correlation Analysis (CCA) is a statistical tool used to analyze the relationship between two sets of variables at the same time. It finds the strongest linear combinations between a predictor variable set and a criterion variable set [13].

Assuming two sets of variables:

$$X = (X_1, X_2, \dots, X_p)$$

$$Y = (Y_1, Y_2, \dots, Y_q)$$

CCA creates canonical variates (U and V) as linear combinations of these sets:

$$U = a_1X_1 + a_2X_2 + \cdots + a_pX_p$$

$$V = b_1Y_1 + b_2Y_2 + \cdots + b_qY_q$$

where a_i and b_i are the canonical coefficients. The canonical correlation coefficient (r_c) measures the strength of the relationship between theses two variates [16]:

$$r_c = \frac{\mathrm{Cov}(U,V)}{\sqrt{\mathrm{Var}(U)\mathrm{Var}(V)}}$$

D. Wilk's Lambda

Wilk's Lambda (Λ) evaluates the statistical significance of the canonical functions by measuring the proportion of unexplained variance between the two variable sets [16]:

$$\Lambda = \prod_{i=1}^{s}(1 - r_i^2)$$

Where r_i is the canonical correlation of the i-th function and s is the total number of canonical functions. Smaller Λ values mean a stronger relationship exists between the variable sets.

E. Chi-Square Approximation

To check if Wilks' Lambda is statistically significant, it is transformed into an approximate Chi-Square (χ^2) statistic [13]:

$$\chi^2 = -(n - \frac{1}{2}(p + q + 3)\ln(\lambda)$$

where n is the sample size, p is the number of variables in the first set, and q is the number of variables in the second set. A significant p-value from this Chi-Square test confirms that the canonical correlation is statistically meaningful.

F. Canonical Determination Coefficient

The Canonical Determination Coefficient R_c^2 shows the percentage of shared variance explained by a canonical function, found by simply squaring the canonical correlation coefficient [16]:

$$R_c^2 = r_c^2$$

For example, if a function has a canonical correlation (r_c) of 0.70, squaring it gives an R_c^2 of 0.49, meaning the canonical variates share 49% of their variance.

IV. Methodology

A. Research Design

A quantitative cross-sectional research approach was employed to identify the relationship between Self-Reward Culture and Financial Control among Bina Nusantara University Students. Data were collected through a structured questionnaire distributed online. The study aimed to examine the multivariate relationship between the two constructs using Canonical Correlation Analysis (CCA).

B. Participants and Data Collection

Data were collected through an online survey administered to undergraduate students at Bina Nusantara University. The questionnaire used a six-point Likert scale ranging from 1 (Disagree) to 6 (Agree) and was designed to measure two constructs: Self-Reward Culture and Financial Control. Self-Reward Culture was represented by three dimensions: Frequency of Self-Reward, Emotional Motivation, and Spending Tendency in Self-Reward. Financial Control was represented by three dimensions: Consumptive Behavior, Financial Management, and Financial Awareness. A total of 105 responses were collected using convenience sampling. After data screening and multivariate outlier detection using Mahalanobis Distance, four observations were identified as multivariate outliers and removed from the dataset. Consequently, the final sample used for analysis consisted of 101 respondents.

C. Research Variables and Measurements

Two latent constructs were examined in this study:

1. Self-Reward Culture (X), represented by three dimensions:
   - Frequency of Self-Reward
   - Emotional Motivation
   - Spending Tendency in Self-Reward
2. Financial Control (Y), represented by three dimensions:
   - Consumptive Behavior
   - Financial Management
   - Financial Awareness

Each dimension was measured using five questionnaire items assessed on a Likert scale. In total, the instrument consisted of 30 indicators, with 15 indicators measuring Self-Reward Culture and 15 indicators measuring Financial Control.

D. Measurement Model Evaluation

To evaluate the measurement model, a second-order Confirmatory Factor Analysis (CFA) was conducted. The analysis was performed using the Maximum Likelihood Robust (MLR) estimator. The CFA assessed the relationships between observed indicators, first-order dimensions, and second-order latent constructs.

To evaluate the model adequacy, several goodness-of-fit indices were used, including the Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR).

E. Validity and Reliability Testing

Construct validity was evaluated using Pearson item-total correlation analysis. An indicator was considered valid when it showed a positive and statistically significant correlation with its corresponding construct score.

Reliability was assessed using Cronbach's Alpha coefficient. A dimension was considered reliable when the Cronbach's Alpha value exceeded 0.70, indicating satisfactory internal consistency among indicators.

F. Assumption Testing

1) Multivariate Normality

Multivariate normality was examined using Mardia's test separately for the Self-Reward Culture and Financial Control constructs. Because the initial results indicated violations of multivariate normality, several data treatment procedures were considered.

First, multivariate outliers were identified and removed using Mahalanobis Distance. Second, item parceling was applied by aggregating indicators into dimensional scores. Finally, a Rank-Based Inverse Normal Transformation (INT) was performed on the parceled data. The transformed data satisfied the multivariate normality assumption and were therefore used in subsequent analyses.

2) Multicollinearity

Multicollinearity was assessed using the Variance Inflation Factor (VIF). VIF values were calculated both at the indicator level and at the dimension level. A VIF value below 5.0 indicated the absence of problematic multicollinearity.

G. Canonical Correlation Analysis

Canonical Correlation Analysis (CCA) was used to investigate the multivariate association between Self-Reward Culture and Financial Control. The analysis was conducted using the three dimensions of Self-Reward Culture as the first variable set and the three dimensions of Financial Control as the second variable set.

The analysis produced canonical functions representing the maximum correlation between linear combinations of the two variable sets. The interpretation of the results included:

- Canonical correlation coefficients,
- Wilks' Lambda significance tests,
- Canonical loadings,
- Cross-loadings, and
- Redundancy indices.

The first statistically significant canonical function was considered the primary basis for interpreting the relationship between Self-Reward Culture and Financial Control among university students.

V. Results and Discussion

This part shows and talks about the results gathered from the statistical examinations. The examination starts with checking the measurement model using Confirmatory Factor Analysis (CFA), then it looks at validity, reliability, normality, and multicollinearity evaluations. After that, Canonical Correlation Analysis (CCA) is done to explore the relationship involving Self-Reward Culture and Financial Control. The outcomes are explained in connection to the research goals and related writings.

A. Confirmatory Factor Analysis (CFA)

A second-order Confirmatory Factor Analysis (CFA) was performed to assess the measurement framework of Self-Reward Culture and Financial Control. The framework included two main constructs, each represented by three different aspects. The indicators of how well the model fits are shown in Table I.

TABLE I. Goodness-of-Fit Indices

Fit Index	Value	Recommended Threshold	Interpretation
CFI	0.863	> 0.90	Marginal Fit
TLI	0.851	> 0.90	Marginal Fit
RMSEA	0.085	< 0.08	Marginal Fit
SRMR	0.169	< 0.08	Poor Fit

According to Table I, the results from the CFA show that the model had a slightly acceptable fit. Even though a few indices fell short of the suggested limits, the factor loadings mostly backed up the connections between the indicators and their related dimensions. As a result, the measurement model was seen as suitable for additional analysis.

B. Validity Test

Validity examination was carried out by employing Pearson correlation analysis to compare each item with its total score. The findings are presented in Table II.

TABLE II. Summary of Validity Test Results

Dimension	Minimum r-value	Maximum r-value	Status
SelfReward_D1	0.825	0.882	Valid
SelfReward_D2	0.843	0.890	Valid
SelfReward_D3	0.723	0.853	Valid
FinancialControl_D1	0.598	0.899	Valid
FinancialControl_D2	0.670	0.811	Valid
FinancialControl_D3	0.719	0.817	Valid

As demonstrated in Table II, all measurements showed positive and significant relationships, which means that every indicator effectively reflects what it is supposed to measure. The indicators measuring Frequency of Self-Reward, Emotional Motivation, and Spending Tendency in Self-Reward effectively capture various aspects of Self-Reward Culture among university students. Similarly, the indicators associated with Consumptive Behavior, Financial Management, and Financial Awareness adequately reflect the construct of Financial Control. These findings suggest that the questionnaire items are appropriate for examining the

relationship between Self-Reward Culture and Financial Control.

C. Reliability Test

Reliability was assessed through Cronbach's Alpha. The findings can be seen in Table III.

TABLE III. RELIABILITY TEST RESULTS

Dimension	Cronbach's Alpha	Interpretation
SelfReward_D1	0.892	Reliable
SelfReward_D2	0.919	Reliable
SelfReward_D3	0.867	Reliable
FinancialControl_D1	0.843	Reliable
FinancialControl_D2	0.783	Reliable
FinancialControl_D3	0.836	Reliable

Every aspect reached Cronbach's Alpha scores higher than 0.70, showing good internal reliability. The results suggest that the indicators used to measure Self-Reward Culture consistently capture students' tendencies to engage in self-reward activities, whether through habitual self-appreciation, emotional motivations, or spending-oriented rewards. Likewise, the indicators measuring Financial Control consistently reflect students' financial behaviors, including consumptive tendencies, financial management practices, and financial awareness. Therefore, the instrument can be considered reliable for investigating the relationship between the two constructs.

D. Multivariate Normality Test

Multivariate normality was assessed using Mardia's test. The findings can be seen in Table III.

TABLE IV. INITIAL MULTIVARIATE NORMALITY TEST

Variable	Skewness p-value	Kurtosis p-value	Results
Self-Reward Culture	< 0.001	< 0.001	**Not Normal**
Financial Control	< 0.001	< 0.001	**Not Normal**

The initial multivariate normality test revealed that both Self-Reward Culture and Financial Control did not satisfy the assumption of multivariate normality. This result indicates that students exhibited considerable variation in their self-reward practices and financial control behaviors. Such heterogeneity is expected because individuals may differ substantially in how frequently they engage in self-reward activities and how effectively they manage their finances. Consequently, additional data treatment procedures were necessary before examining the multivariate relationship between the two constructs.

E. Handling Violation of Normality

1) Outlier Removal

Mahalanobis Distance was employed to detect multivariate outliers. Four observations were identified and removed.

TABLE V. OUTLIER REMOVAL SUMMARY

Description	Number of Obesrvation
Initial Sample Size	105
Outliers Removed	4
Final Sample Size	101

Despite the removal of outliers, the assumption of multivariate normality was still not satisfied.

2) Parceling and Transformation

Item parceling was applied by aggregating indicators into dimensional scores. Subsequently, Rank-Based Inverse Normal Transformation (INT) was performed.

TABLE VI. MULTIVARIATE NORMALITY AFTER PARCELING AND TRANSFORMATION

Construct	Skewness p-value	Kurtosis p-value	Results
Self-Reward Culture	0.059	0.352	**Normal**
Financial Control	0.554	0.597	**Normal**

After transformation, both constructs satisfied the assumption of multivariate normality and were considered appropriate for Canonical Correlation Analysis.

F. Multicollinearity Test

Multicollinearity was evaluated by means of the Variance Inflation Factor (VIF). The findings are shown in Table VII.

TABLE VII. SUMMARY OF MULTICOLLINEARITY

Construct	Maximum VIF	Interpretation
Self-Reward Culture	4.327	No Multicollinearity
Financial Control	2.469	No Multicollinearity

Since all the VIF values were under the threshold of 5.0, multicollinearity was not considered problematic.

G. Canonical Correlation Analysis

1) Canonical Correlation Coefficients

Canonical Correlation Analysis was performed to explore how Self-Reward Culture relates to Financial Control.

TABLE VIII. CANONICAL CORELATION COEFFICIENTS

Canonical Function	Correlation
Function 1	0.663
Function 2	0.345
Function 3	0.029

The initial canonical function showed the most significant connection with a canonical correlation coefficient of 0.663. This finding suggests a fairly strong link between Self-Reward Culture and Financial Control among college students. In practical terms, variations in students' self-reward tendencies are associated with variations in their financial control characteristics. Students who demonstrate stronger self-reward behaviors tend to exhibit distinct patterns of financial control, particularly in relation to spending behavior and financial decision-making.

2) Significance Test of Canonical Functions

The significance of each canonical function was evaluated using Wilks' Lambda.

TABLE IX. WILKS' LAMBDA SIGNIFICANCE TEST

Tested Functions	p-value	Interpretation
Functions 1–3	< 0.001	Significant
Functions 2–3	0.015	Significant
Function 3	0.779	Not Significant

The first canonical function was statistically significant and demonstrated the strongest explanatory power.

3) Canonical Loadings

Canonical loadings were analyzed to determine the contribution of each dimension.

TABLE X. CANONICAL LOADINGS OF SELF-REWARD CULTURE

Dimension	Loading
Frequency of Self-Reward	-0.740
Emotional Motivation	- 0.895
Spending Tendency in Self-Reward	-0.940

TABLE XI. CANONICAL LOADINGS OF FINANCIAL CONTROL

Dimension	Loading
Consumptive Behavior	- 0.975
Financial Management	- -0.271
Financial Awareness	-0.210

The canonical loading analysis shows that Spending Tendency in Self-Reward is the most influential dimension within Self-Reward Culture, followed by Emotional Motivation. This finding suggests that the financial aspect of self-reward, particularly the tendency to spend money as a form of self-appreciation, plays a central role in explaining the construct among university students.

On the Financial Management aspect, Consumer Habits stood out as the most significant factor. This finding shows that the link between a Self-Reward Culture and Financial Management mainly shows through how students choose to spend money and their buying choices, rather than just through understanding finances or managing money alone.

4) Cross Loadings

Cross loadings were calculated to examine the relationship between each dimension and the opposite canonical variate.

TABLE XII. CROSS LOADINGS

Dimension	Cross Loading
Frequency of Self-Reward	-0.490
Emotional Motivation	-0.593
Spending Tendency in Self-Reward	-0.623
Consumptive Behavior	-0.646
Financial Management	-0.180
Financial Awareness	-0.139

The highest cross loadings were observed for Spending Tendency in Self-Reward and Consumptive Behavior, indicating that these dimensions primarily drive the relationship between the two constructs.

5) Redundancy Analysis

A review of redundancy was done to find out how much variation each main function accounts for.

TABLE XIII. REDUNDANCY ANALYSIS

Function	Self-Reward	Financial Control
Function 1	0.326	0.156
Function 2	0.013	0.016
Function 3	0.000	0.000

The analysis of redundancy shows that the first main function accounts for 32.6% of the differences in Self-Reward Culture and 15.6% of the differences in Financial Control. These results imply that the complex connections found in this research are significant, especially in clarifying how aspects of self-reward actions relate to changes in students' financial control traits. The much smaller redundancy values for the second and third functions further support the idea that the first main function illustrates the key link between the two concepts.

H. Discussion

The research shows an important link between a culture of self-reward and financial management among college students. The initial canonical function produced a canonical correlation coefficient of 0.663, which is statistically meaningful ($p < 0.001$), indicating that there is a fairly strong connection between these two concepts. This finding implies that the ways students reward themselves are linked to their financial management skills.

The canonical loading analysis identified Spending Tendency in Self-Reward as the strongest contributor within the Self-Reward Culture construct, while Consumptive Behavior emerged as the dominant dimension within Financial Control. This discovery suggests that the connection between the two ideas mainly comes from activities where people reward themselves by spending money and the way students buy things. In other words, self-reward behavior becomes most influential when it involves financial expenditure, which subsequently relates to students' tendencies to make purchases for personal satisfaction.

The cross-loading results further support this interpretation. Students who frequently engage in self-reward activities involving financial expenditure tend to exhibit stronger consumptive tendencies. This pattern may occur because self-reward often serves as a psychological mechanism for providing immediate gratification after completing responsibilities, achieving goals, or coping with emotional stress. As a result, spending is frequently perceived as a justified form of self-appreciation rather than merely a financial transaction.

These findings are consistent with previous studies suggesting that self-reward behavior is associated with consumptive behavior among Generation Z. Prior research argues that self-reward frequently functions as a psychological justification for discretionary spending and impulse purchases. The present study extends these findings by demonstrating that self-reward culture is not only related to purchasing behavior but is also significantly associated

with broader aspects of financial control among university students.

Furthermore, the dominance of Consumptive Behavior compared to Financial Management and Financial Awareness suggests that the influence of self-reward culture is more visible in students' spending decisions than in their financial planning or awareness. This finding implies that although students may understand the importance of managing their finances, self-reward motivations can still encourage consumption-oriented behavior when emotional or psychological needs become more salient.

Overall, the results demonstrate that Self-Reward Culture and Financial Control are significantly interconnected. The dominance of Spending Tendency in Self-Reward and Consumptive Behavior highlights the importance of balancing self-appreciation practices with responsible financial management. Therefore, financial education programs targeting university students should not only emphasize financial literacy but also address the psychological motivations underlying spending behavior. By understanding the role of self-reward in financial decision-making, students may be better equipped to maintain healthy financial habits while still engaging in positive forms of self-appreciation.

VI. Conclusion

This study examined the relationship between self-reward culture and financial control among university students using a multivariate statistical approach. Confirmatory Factor Analysis (CFA) validated our measurement model and Canonical Correlation Analysis (CCA) was applied to analyze the specific relationships between the different dimensions of self-reward culture and financial control.

The results show that self-reward behavior significantly impacts how students manage their money. While treating oneself can be an emotional satisfaction and personal motivation, overindulging in reward-based spending can undermine financial discipline. Ultimately, these results emphasize that students need to find a health balance between immediate spending and long-term responsible financial choices.

By providing empirical evidence on this relationship, this study adds a new perspective to the existing literature on consumer behavior and financial management. This study offers a targeted look at how self-reward mindsets specifically impact financial control.

For future research, this framework could be expanded by adding variables like financial literacy, self control, or social media influence. Additionally, testing larger and more diverse student samples would help generalize these insights for the emerging generation.

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